

Aasheesh Bajpai, Ph.D.

✉ aasheesh@iitk.ac.in  <https://sites.google.com/view/aasheeshbajpai/home>
 <https://www.linkedin.com/in/aasheeshbajpai/>
 <https://home.iitk.ac.in/~aasheesh/>



Education

- 2020 – Present  **Ph.D., Aerospace Engineering, Indian Institute of Technology Kanpur** at NFSL.
(CPI : 10/10)
Thesis title: *Computational modeling of dusty gas flow in DSMC-DEM framework for planetary landing.*
Thesis supervisor: *Prof. Rakesh Kumar*
- 2017 – 2019  **M.Tech. Cryogenic Engineering, Institute of Technology Kharagpur** at PED Lab.
(CPI : 8.51/10)
Thesis title: *Development of test rig towards experimental investigation of Reverse Brayton Cryocooler used for cooling of High Temperature Superconductors.*
Thesis supervisor: *Prof. Parthasarathi Ghosh*
- 2012 – 2016  **B.Tech. Mechanical Engineering, AKTU, U.P.** at AKGEC, Ghaziabad. (Honours)
Thesis title: *Design, Fabrication and Review of GO-KART.*
Responsibility: Managed a team of 10 engineers
Thesis supervisor: *Mr. Ankur Tyagi*

Work experience

- July – Dec 2019  **Research Assistant** at IIT Kharagpur
I worked on enhancing exergetic efficiency of hydrogen liquefaction plant in which specifically we enhanced the efficiency of cryogenic turboexpanders to increase the overall efficiency of process.

Training

- 17 – 20 May 2022  **High Performance Computing (HPC) for Computational Fluid Dynamics (CFD) Applications** Organised by IIT Bombay and C-DAC under the aegis of National Supercomputing Mission(NSM)
- July – August 2015  **Vocational Training in NTPC, Dadri**
Work: *Boiler maintenance and Ash handling department, Learnt about Power plant process, Rankine cycle and various equipment used etc. Also submitted a project report on Cooling Tower*
- June – July 2014  **In workshop training Indian Railways workshop, Izzatnagar, Bareilly**
Work: *Braking system in locomotive (Air brakes, pneumatic brakes), working of engine of locomotive, carriage maintenance.*

Research Publications

Journal Articles

-  1. A. Appar, A. Bajpai, U. Shivkumar, S. Naspoori, and R. Kumar, "Rarefied gas effects on hypersonic flow over a transpiration-cooled flat plate," *Submitted to 'Physics of Fluids'*, 2022.

- 2 A. Bajpai, A. Bhateja, and R. Kumar, "Plume surface interaction during lunar landing," *Manuscript under preperation*, 2022.
- 3 A. Bajpai, R. Dutta, and P. Ghosh, "Parameter estimation of equipment for development of an experimental setup of a reverse brayton cryocooler for cooling hts cables," *Indian Journal of Cryogenics*, vol. 44, p. 83, Jan. 2019.  DOI: 10.5958/2349-2120.2019.00014.1.

Conference Proceedings

- 1 A. Bajpai, A. K. Dhillon, R. Dutta, and P. Ghosh, "Root cause analysis of early performance deterioration of an existing helium liquefier using process simulation," in *15th Cryogenics, IIR international conference proceedings*.
- 2 A. K. Dhillon, A. Bajpai, R. Dutta, and P. Ghosh, "The effect of hts heat rejection conditions on performance of reverse brayton cryocooler," in *15th Cryogenics, IIR international conference proceedings*.

Conference Presentation

- 1 A. Appar, U. Shivkumar, A. Bajpai, S. Naspoori, and R. Kumar, *Validity of the blowing correction correlation in rarefied flow regimes*, 32nd International Symposium on Rarefied Gas Dynamics (RGD32) at Seoul, South Korea on 4th to 8th July, 2022.
- 2 A. Bajpai, A. Appar, S. Naspoori, and R. Kumar, *Numerical investigation of pyrolysis gas interaction with hypersonic reentry flow-field*, 32nd International Symposium on Rarefied Gas Dynamics (RGD32) at Seoul, South Korea on 4th to 8th July, 2022.
- 3 A. Bajpai, M. Bhavsar, D. Dutta, A. Bhateja, and R. Kumar, *Lagrangian-lagrangian simulation of dusty gas flow past a cylinder*, Traffic & Granular Flow (TGF-2022)- at IIT Delhi, Delhi, India on 15th to 17th October, 2022.
- 4 M. Bhavsar, A. Bajpai, A. Khan, and R. Kumar, *Granular flow past an elliptical obstacle*, Traffic & Granular Flow (TGF-2022)- at IIT Delhi, Delhi, India on 15th to 17th October, 2022.
- 5 R. Kumar, A. Chinappan, A. Bajpai, A. Bhateja, and M. Bhavsar, *Modeling of lunar dust dispersion using two-way coupled lagrangian-lagrangian framework*, 32nd International Symposium on Rarefied Gas Dynamics (RGD32) at Seoul, South Korea on 4th to 8th July, 2022.
- 6 A. Bajpai, A. K. Dhillon, R. Dutta, and P. Ghosh, *Root cause analysis of early performance deterioration of an existing helium liquefier using process simulation*, IIR international conference, April 7-11, 2019, Prague, Czech Republic, Europe, 2019.
- 7 A. Bajpai, R. Dutta, and P. Ghosh, *Parameter estimation of equipment for development of an experimental setup of a reverse brayton cryocooler for cooling high temperature superconducting cables*, National Symposium in Cryogenics and Superconductivity-27 at IIT Bombay, Mumbai, India on 16th to 18th January, 2019.

Poster presentations

- 1 A. Bajpai, A. Appar, S. Naspoori, and R. Kumar, *Numerical investigation of pyrolysis gas interaction with hypersonic reentry flow-field*, 32nd International Symposium on Rarefied Gas Dynamics (RGD32) - 2022 at Seoul, South Korea on 4th to 8th July, 2022.
- 2 A. Bajpai, A. K. Dhillon, and P. Ghosh, *The effect of hts heat rejection conditions on performance of reverse brayton cryocooler*, IIR international conference, April 7-11, 2019, Prague, Czech Republic, Europe, 2019.

Skills

Languages	Strong reading, writing and speaking competencies for English and Hindi .
Coding	Modern Fortran, C, C++, Python, Linux BASH Programming, $\LaTeX$
Python libraries	Numpy, Pandas
Parallel programming	OpenMP, MPI.
Operating Systems	Linux(Ubuntu, Red Hat Enterprise), Windows, MacOS
HPC and DevOps Skills	Comfortable with SLURM, PBS Pro, Software compilation/building from source on Linux platform, and benchmarking
Simulation Techniques	Direct Simulation Monte Carlo (DSMC), Discrete Element Method (DEM), Computational Fluid Dynamics (CFD) (FDM)
Softwares	SPARTA(DSMC), LIGGGHTS(DEM), OpenFOAM(CFD), CFD-DEM, SOLIDWORKS, Pro-E, Ansys Fluent, ANSYS CFX, BladeGen.
Other softwares	MS Office, Visual Studio, Tecplot, Ovito, LabVIEW, VEUSZ, ORIGIN, Engineering Equation Solver, Aspen- HYSYS, Aspen-EDR, HYSYS- Dynamics.
Web Dev	HTML, CSS
Misc.	Academic research, Strong analytical and problem-solving skills, literature survey, technical writing, teaching, training, consultation, $\LaTeX$ typesetting and publishing, story-telling and presentation skills.

Miscellaneous Experience

Awards and Achievements

- 2019  **Got the prestigious Prime Minister Research Fellowship (PMRF)** from Ministry of Education, Govt. of India for Ph.D. program.
- 2022  **Achieved full financial assistance from IIT Kanpur, to attend the RGD-32, International Symposium on Rarefied Gas Dynamics** at Seoul, South Korea on 4th to 8th July, 2022.
- 2019  **Achieved full financial assistance from IIT Kharagpur under the category of Best Academic International Conference, to attend the 15th Cryogenics, IIR international conference** at Prague, Czech Republic, Europe. from April 7-11, 2019.
- 2015  **Participated and won first runner-up title in IGC LPU and managed a team of 10 engineering students**, Lovely Professional University, Haryana.

Teaching Experience

- Fall 2022  **Tutor NPTEL.** Computational Fluid Dynamics and Heat Transfer.
Took weekly live classes for 2188 Students
- Spring 2022  **Tutor NPTEL.** Introduction to programming with C.
Took weekly live classes for 44925 Students
- Fall 2021  **Teacher.** Introduction to Thermodynamics
Took recorded classes for 40 Students.
-  **Teaching Assistant, IIT Kanpur.**
AE451(A) : Experiments in Fluid (Aerodynamics Experiments)

Miscellaneous Experience (continued)

- Spring 2021  **Teaching Assistant, IIT Kanpur.**
(ESO201A: Thermodynamics)
- Spring 2019  **Teaching Assistant, IIT Kharagpur.**
(CR61022: Cryogenic Expansion Devices (Turbomachinery))
- Fall 2018  **Teaching Assistant, IIT Kharagpur.**
(CR61011: Compressors and Pumps)

Certification

- January-2015  **Automation technologies-Basic level.** Awarded by 'Rexroth Bosch Group'.
Topics covered-Industrial Hydraulics, Industrial pneumatics, PLC
- 2014-15  **Training program on Pro/Engineer .** Awarded by 'RICC AKGEC'.

Cocurricular and extracurricular activities

- Leadership and mentoring experience  Assisted 2 Bachelor's project and 1 B.tech internship,
Played a vital role in preparing the course curriculum for the
course 'Cryogenic Expansion Devices', at IIT Kharagpur
- Social work  Member of counselling services, IIT Kanpur,
Active member of hall task force for tackling Covid-19 issues,
Served as a member of Mess committee of hall.
- Membership  Former head of Indian Society for Technical Education at
AKGEC.
- Managerial skills  Organized a successful fest at undergraduate level, Organized
research scholar day at IIT Kharagpur, India
- Fitness skills  Active practitioner of yoga, pranayama, meditation and a cy-
cling enthusiast.
- Interests  Space science & astronomy, rocket science , international
affairs and geopolitics, cooking, ayurveda, cricket, Advaita
Vedanta philosophy, music and travelling.

References

Available on Request